



7717 - Topological Mapping of Superbubbles and ISM Structures with JWST

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Feedback-driven superbubbles are known to influence the progression of star formation and the evolution of the interstellar medium (ISM) on galactic scales. JWST has transformed our ability to resolve the precise geometry and local environments of superbubbles in nearby galaxies, but current methods for the identification of candidate bubbles are not easily scaleable and struggle to simultaneously map both superbubbles and their local environments (such as molecular clouds and filaments) in a self-consistent framework. We propose an archival program to develop and release an open-source software package for the topological analysis of structures in astronomical images. We will apply this method to a sample of 81 galaxies observed by JWST, and map the structure of 100,000 superbubbles, molecular clouds, and filaments. This analysis will reveal how superbubbles influence the interstellar medium and star formation across diverse galactic environments.

OBSERVING DESCRIPTION